

Lecture Notes no 10

Statistical Analysis of the Two Way ANOVA - Fixed Effects Model

The following notation will be used here: Let $y_{i..}$ denote the total of all observations for the i th level of factor A

$$y_{i..} = \sum_{j=1}^b \sum_{k=1}^n y_{ijk},$$

$y_{.j.}$ denote the total of all observations for the j th level of factor B

$$y_{.j.} = \sum_{i=1}^a \sum_{k=1}^n y_{ijk},$$

$y_{ij.}$ denote the total of all observations in the ij th cell,

$$y_{ij.} = \sum_{k=1}^n y_{ijk},$$

and $y_{...}$ denotes the grand total of all observations

$$y_{...} = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n y_{ijk}.$$

The corresponding factor A , factor B , cell and grand sample means are given by:

$$\bar{y}_{i..}, \bar{y}_{.j.}, \bar{y}_{ij.}, \text{ and } \bar{y}_{...}$$

respectively.

The total corrected sum of squares identity holds for this design as well:

$$\begin{aligned} \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (y_{ijk} - \bar{y}_{...})^2 &= bn \sum_{i=1}^a (\bar{y}_{i..} - \bar{y}_{...})^2 + an \sum_{j=1}^b (\bar{y}_{.j.} - \bar{y}_{...})^2 \\ &\quad + n \sum_{i=1}^a \sum_{j=1}^b (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2 \\ &\quad + \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (y_{ijk} - \bar{y}_{ij.})^2. \end{aligned}$$

The equation above is referred to as the decomposition of the variability in this two way factorial model:

$$SS_T = SS_A + SS_B + SS_{AB} + SS_E.$$

The associated expected mean squares are given by:

$$E(MS_A) = E\left(\frac{SS_A}{a-1}\right) = \sigma^2 + \frac{bn \sum_{i=1}^a A_i^2}{a-1},$$

$$E(MS_B) = E\left(\frac{SS_B}{b-1}\right) = \sigma^2 + \frac{an \sum_{j=1}^b B_j^2}{b-1},$$

$$E(MS_{AB}) = E\left[\frac{SS_{AB}}{(a-1)(b-1)}\right] = \sigma^2 + \frac{n \sum_{i=1}^a \sum_{j=1}^b (AB_{ij})^2}{(a-1)(b-1)},$$

and

$$E(MS_E) = E\left[\frac{SS_E}{ab(n-1)}\right] = \sigma^2.$$

Testing Hypothesis for a Two Way ANOVA - Fixed effects model

The model here is:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{k(ij)}, \quad 1 \leq i \leq a, 1 \leq j \leq b, 1 \leq k \leq n.$$

For this model we are interested in testing simultaneously the following three null hypotheses:

$$H_{01} : A_i = 0, 1 \leq i \leq a, \quad (1)$$

$$H_{02} : B_j = 0, 1 \leq j \leq b, \quad (2)$$

and

$$H_{03} : (AB)_{ij} = 0, 1 \leq i \leq a, 1 \leq j \leq b. \quad (3)$$

One can show that the test statistic for testing the null hypothesis $H_{01} : A_i = 0, 1 \leq i \leq a$ is:

$$F_1 = \frac{MS_A}{MS_E},$$

the rest statistic for testing $H_{02} : B_j = 0, 1 \leq j \leq b$ is:

$$F_2 = \frac{MS_B}{MS_E},$$

and the test statistic for testing $H_{03} : (AB)_{ij} = 0, 1 \leq i \leq a, 1 \leq j \leq b$ is:

$$F_3 = \frac{MS_{AB}}{MS_E}.$$

Based on the expected mean squares, presented on page 2, one can show that each of the null hypotheses stated above are rejected for large value of the computed F statistics.

Since we are testing the above three null hypotheses simultaneously and the test statistics that we are using are not independent of each other, we will use the following Bonferroni type correction. If we set the over all significance level for testing the above three null hypothesis to be equal to α , then we will reject the null hypotheses that have a p-value $< \alpha/3$. For example, if we want to have an overall significance level for testing the above three null hypothesis to be equal to .05, then we will reject the null hypotheses that have a p-value $< .05/3 = .0166$. The general rule is: we compare the p-value corresponding to testing each null hypothesis to a *nominal* alpha level, defined as follows:

$$\text{nominal } \alpha \text{ level} = \frac{\text{overall } \alpha \text{ level}}{\text{number of hypotheses tested}}.$$

The ANOVA table for a fixed effects two factorial design is given by

Source of Variability	SS	df	MS	F
<i>A</i> treatments	SS_A	$a - 1$	MS_A	$\frac{MS_A}{MS_E}$
<i>B</i> treatments	SS_B	$b - 1$	MS_B	$\frac{MS_B}{MS_E}$
<i>AB</i> interactions	SS_{AB}	$(a - 1)(b - 1)$	MS_{AB}	$\frac{MS_{AB}}{MS_E}$
Error	SS_E	$ab(n - 1)$	MS_E	
Total	SS_T	$abn - 1$		

Checking the Validity of the Underlying Assumption for the Model

Here too the main diagnostic tool is residual analysis. The residuals for this model are

$$e_{ijk} = Y_{ijk} - \bar{Y}_{ij}.$$

We are going to examine:

- the normal probability plot of the residuals.
- the plot of the residuals vs the predicted values $\hat{Y}_{ijk} = \bar{Y}_{ij}$.
- The p-value of the W statistic in PROC UNIVARIATE.
- The size of the RStudent residuals and Cook's D values

Multiple Comparisons

If only one or both of the main effects (*A* or *B*) are significant, then we will use Tukey's multiple comparison procedure to compare the means of the significant main effect(s).

If only the interaction effect is significant we will use Tukey's multiple comparison procedure to compare the *ab* means of the interaction effect.

If one or both main effects are significant and the interaction effect is significant, we will use Tukey's multiple comparison procedure to compare the means displayed vertically in the interaction plot(s) of the significant main effect vs the other main effect.